Trust and Compliance-Aware Decentralised Emergency Vehicle Corridor Coordination System Using Local Digital Twins and Swarm Intelligence

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Abstract. Emergency vehicles can be delayed by congestion, uncoordinated yielding and unreliable traffic information. This paper presents a trust- and compliance-aware framework for decentralised emergency corridor coordination. Participating vehicles are represented as independent virtual agents that exchange signed state messages and maintain local digital twins of nearby traffic. PKI-based credentials and ECDSA signatures support identity and message-integrity checks, while timestamps and sequence numbers provide freshness checks. A behavioural trust score and Isolation Forest assess the reliability and abnormality of received information. LSTM-based trajectory prediction and XGBoost-based compliance estimation inform Distributed Particle Swarm Optimisation (DPSO) for local yielding coordination; A* supports emergency route planning. Observed driver responses feed an event-triggered replanning loop subject to safety constraints. The proposed software prototype is intended for evaluation in a Python-generated synthetic traffic environment, with a React dashboard for inspection. This paper specifies the architecture and evaluation protocol; quantitative performance results are not reported because validated experimental measurements were not supplied.
Keywords: Emergency vehicle coordination; decentralised V2X; local digital twin; behavioural trust; anomaly detection; trajectory prediction; driver compliance; DPSO; A*.

1 Introduction

Emergency response time is a critical operational factor for ambulance, fire and police services. In dense urban traffic, an emergency vehicle may encounter queues, blocked lanes, inconsistent vehicle-state reports and drivers who do not respond as expected to yielding requests. Conventional route selection alone cannot resolve these local interactions: a route may be topologically short yet temporarily impassable, and a corridor plan can fail when neighbouring vehicles do not comply.

Connected-vehicle communication and digital-twin methods provide mechanisms for sharing state and reasoning about traffic dynamics. However, a system that relies on a single global traffic representation can introduce communication and coordination dependencies. The present project instead considers a set of virtual vehicle agents, each maintaining a local digital twin of nearby traffic and making decisions from locally received information. The design integrates message authentication, behavioural trust, anomaly screening, prediction, decentralised coordination and route replanning.

The contribution is an integrated system design rather than a claim of a new individual learning or optimisation algorithm. Specifically, the framework (i) separates credential verification from behavioural reliability assessment; (ii) maintains per-agent local traffic representations; (iii) combines trajectory and yielding-compliance estimates with distributed coordination; and (iv) uses observed responses to trigger route and corridor updates. The implementation scope is a Python-based synthetic traffic environment with virtual agents and a React visualisation layer, as specified in the project documentation.

2 Related Work

Digital-twin research in intelligent transportation systems addresses architecture, synchronisation, traffic services and implementation challenges. Ge et al. review digital-twin intelligent transportation systems and identify architectural and service-level considerations across the ITS domain [1]. Recent surveys also examine digital twins for connected and autonomous vehicles, including safety, traffic management and adaptive response [2]. These studies motivate the use of a digital representation for traffic-state reasoning, while leaving design choices about the distribution and authority of individual twins to the application.

Emergency lane-clearing research has considered cooperative movement and route-level optimisation. The two-stage lane pre-clearing approach of Wu et al. formulates cooperative driving around emergency vehicles while balancing emergency progress and disruption to surrounding connected vehicles [3]. More recent work studies distributed learning for lane clearance in connected and automated environments [4]. These works demonstrate that corridor formation is a coupled movement problem rather than a route-planning problem alone.

The proposed framework differs in its intended integration boundary: local digital twins are maintained by individual virtual vehicle agents; received data is screened through cryptographic validation and behavioural trust; predicted yielding behaviour is included in the coordination inputs; and observed non-compliance can trigger replanning. This paper does not

claim that these component algorithms are individually novel. The project documentation also lists additional related studies, but complete publication metadata for some entries was not available in the supplied report; those entries are therefore not used as verified references.

3 Problem Formulation and System Model

Let the road network be a directed graph G=(V,E), where V denotes road junctions and E denotes road segments. At simulation time t, vehicle i has state x_i(t)=[p_i(t),v_i(t),a_i(t),h_i(t),l_i(t)], representing position, speed, acceleration, heading and lane or movement information. An emergency vehicle e has a destination d_e, priority q_e and current route r_e(t). Each ordinary vehicle is represented by an independent agent with a communication neighbourhood N_i(t), determined by the configured local communication range.

The objective is to generate a feasible emergency corridor while limiting disruption and maintaining safety. A generic coordination objective can be expressed as minimising $J = \alpha T_e + \beta D_n + \gamma R_s + \delta C_m + \varepsilon N_{nc}$, where T_e is emergency travel time, D_n is aggregate delay to normal vehicles, R_s is a safety-risk penalty, C_m is communication cost and N_nc is the number or weighted severity of non-compliant responses. The coefficients are design parameters that must be selected and tested; no numerical values are asserted here. Collision avoidance, road-boundary constraints and legal movement constraints are hard constraints, not merely soft penalties.

The system assumes that virtual agents have unique identities and public-key credentials provisioned by a project-defined certificate authority. It also assumes that message exchange is available within a simulated neighbourhood and that synthetic vehicle trajectories are generated at a consistent time step. These assumptions are suitable for a software prototype but do not establish deployment readiness for real vehicular networks.

4 System Architecture

Figure 1 summarises the proposed data and decision flow. A Python traffic generator creates road states, vehicle agents and emergency scenarios. Agents broadcast signed messages containing vehicle identity, timestamp, sequence number and state fields. Receivers validate credentials, signatures and freshness before using the message. Behavioural trust and anomaly checks then determine whether the received state should update the local twin, be down-weighted or be excluded pending further evidence.

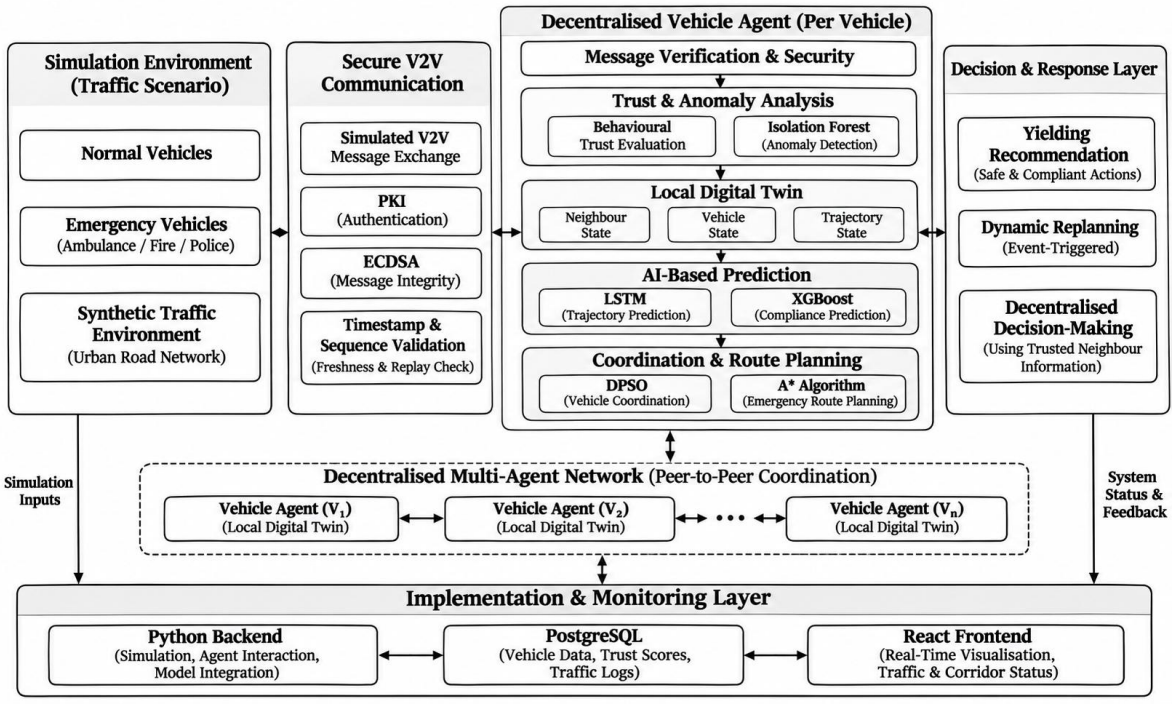


Fig. 1. Proposed decentralised emergency-corridor coordination workflow.

Each local digital twin stores a bounded view of nearby vehicles and the latest accepted state estimates. The prediction layer estimates future motion and the probability that a driver will follow a yielding recommendation. DPSO uses local state, predicted trajectories and compliance estimates to coordinate feasible yielding actions. A* searches the road graph for a route using travel-time and operational costs. The dashboard visualises local traffic, emergency priority, trust status, recommendations and replanning events; it is not the central authority for vehicle movement decisions.

Based on the coordination outcomes, each vehicle agent evaluates the feasibility of the recommended actions while considering collision avoidance, lane constraints and traffic conditions. The resulting decisions are shared with neighboring agents to maintain coordinated movement and support emergency corridor

formation. Vehicle movements and driver responses are continuously monitored to identify deviations from the recommended actions. If non-compliance or significant traffic changes are detected, the affected agents update their local digital twins and initiate a new coordination and routing cycle.

5 Security and Trust Management

5.1 Message authentication and freshness

A message m_i is associated with a sender certificate and a digital signature σ_i=Sign_sk_i(H(m_i)), where sk_i is the sender's private key and H is a cryptographic hash. A receiver checks the certificate chain and verifies σ_i using the corresponding public key. It also checks timestamp age and sequence-number progression to reject stale or replayed messages. ECDSA provides signature authenticity and integrity when correctly implemented; it does not establish that a correctly signed position or speed report is factually accurate.

5.2 Behavioural trust and anomaly screening

To assess reported-state reliability, the design uses a weighted trust score T_i=w_pP_i+w_cC_i+w_fF_i+w_sS_i, where P_i represents position plausibility, C_i movement consistency, F_i message freshness and S_i consistency with independently received neighbour information. The component scores are normalised to a common range and the weights sum to one. The project must specify the normalisation functions, thresholds and weight-selection procedure before experimental evaluation.

Isolation Forest is applied to feature vectors derived from reported movement and contextual state. An anomaly score indicates deviation from learned normal patterns; it is not, by itself, proof of malicious intent. Trust and anomaly outputs should therefore be combined with authentication status and temporal evidence. A valid signature with an anomalous trajectory remains an authenticated but potentially unreliable report.

6 Prediction and Decentralised Coordination

6.1 Local digital twin and trajectory prediction

For each agent i, the local digital twin maintains a time-indexed set of nearby vehicle states, road-segment context, emergency requests and message-quality metadata. Accepted updates are reconciled using timestamps and trust weights. An LSTM model receives a sequence of historical states and predicts future positions or movement states over a selected horizon. The prediction target, sampling interval, forecast horizon and input window must be fixed during dataset construction and reported with the evaluation.

6.2 Driver compliance estimation

XGBoost is used to estimate P(y_i=1|z_i), where y_i indicates whether vehicle i complies with a yielding recommendation and z_i contains behavioural and traffic features. Candidate features include distance to the emergency vehicle, relative speed, lane availability, congestion, previous compliance and the recommendation type. The estimate is a probability, not a guarantee that a driver will yield. The coordination layer should retain a conservative fallback when prediction confidence is low.

6.3 DPSO-based movement coordination

DPSO is intended to coordinate local yielding decisions among agents using their neighbourhood information. A particle may encode a candidate set or sequence of yielding actions; its objective evaluates emergency progress, normal-traffic delay, safety risk, communication burden and predicted non-compliance. Each agent updates its local candidate using its own best-known solution and information exchanged with neighbouring agents. The implementation must preserve this distributed update structure; a single central optimiser controlling every vehicle would not constitute DPSO-based decentralised coordination.

The optimisation output is treated as a recommendation subject to safety and feasibility checks. A vehicle should not execute a movement that violates collision-avoidance, lane geometry or right-of-way constraints, even if the movement reduces the objective value.

7 Route Planning and Compliance-Aware Replanning

A* operates on the road graph to identify a feasible emergency route. For node n, its evaluation function is f(n)=g(n)+h(n), where g(n) is accumulated route cost and h(n) is an admissible estimate of remaining cost when optimality is required. Edge costs can reflect travel time, congestion and operational restrictions. A* selects the route; DPSO coordinates local vehicle yielding actions along the route. These functions are complementary and should not be conflated.

After recommendations are issued, the system observes the subsequent synthetic vehicle states. If an agent does not yield within a configured response interval, or if a lane becomes blocked or traffic conditions materially change, the affected local twins are updated and the coordination process is repeated. DPSO may revise yielding actions and A* may select an alternative route. Replanning should be event-triggered and bounded to avoid unstable oscillation between routes.

8 Experimental Design and Evaluation Protocol

The project documentation specifies synthetic vehicle-state, emergency-vehicle, driver-behaviour and AI-

training data. The traffic generator should produce continuous, physically plausible trajectories with consistent time steps, lane constraints and configurable congestion. Driver profiles should include cooperative and non-cooperative behaviour. Training, validation and test partitions should be separated by time or scenario so that near-duplicate trajectory windows do not leak across partitions.

The planned evaluation should include at least: (i) normal traffic with cooperative drivers; (ii) heavy congestion; (iii) partial and complete yielding non-compliance; (iv) stale, replayed or inconsistent messages; and (v) combined congestion and communication disturbances. Baselines should include route planning without coordination, coordination without compliance prediction, and the complete proposed pipeline. Ablation studies should remove trust filtering, trajectory prediction or compliance estimation one at a time to isolate their effects.

Table 1. Planned evaluation scenarios and measures.

Scenario	Perturbation	Primary measures
Cooperative flow	Moderate demand; drivers yield	Emergency travel time; corridor formation time
Congestion	High density and blocked segments	Emergency delay; normal-vehicle delay
Non-compliance	Selected drivers ignore recommendations	Compliance F1; replanning response time
Message faults	Stale, replayed or inconsistent state	Replay rejection; anomaly and trust behaviour
Combined stress	Congestion, non-compliance and message faults	Safety conflicts; corridor success; communication overhead

The principal measures are emergency travel time and delay, corridor formation time, corridor completion rate, normal-vehicle delay, LSTM trajectory MAE/RMSE, XGBoost accuracy/precision/recall/F1, authentication and replay rejection counts, anomaly-detection performance, DPSO execution time, communication overhead and unsafe-conflict count. Results must be reported over repeated scenario seeds with means and dispersion, and prediction metrics must be computed on held-out scenarios.

9 Results Status and Discussion

No measured experimental results, trained-model metrics or validated runtime measurements were supplied with the project documentation. Accordingly, this paper reports the proposed architecture and evaluation protocol rather than claiming performance gains. Table 2 is intentionally a results template and must be completed after experiments.

Table 2. Experimental results template; values are pending measurement.

Metric	Baseline	Proposed system	Status
Emergency travel time (s)	[TO ADD]	[TO ADD]	Not measured
Corridor formation time (s)	[TO ADD]	[TO ADD]	Not measured
Compliance F1-score	[TO ADD]	[TO ADD]	Not measured
Trajectory MAE/RMSE	[TO ADD]	[TO ADD]	Not measured
Unsafe conflicts	[TO ADD]	[TO ADD]	Not measured
Communication overhead	[TO ADD]	[TO ADD]	Not measured

The design is expected to make the corridor-generation process more responsive to uncertain vehicle behaviour by combining authenticated messages, reliability assessment and compliance-aware replanning. This is a hypothesis to be tested, not an established result. Synthetic data also limits external validity: generated driver behaviour may not represent the diversity of real road users, and cryptographic checks in a prototype do not establish resilience to all vehicular-network attacks.

10 Limitations and Future Work

The current scope is a software-based prototype using Python-generated synthetic traffic and virtual vehicle agents. It does not include physical vehicles, real V2X radios, roadside infrastructure or live-road deployment. The project documentation identifies Python/FastAPI for backend services, React for visualisation, PostgreSQL for records, scikit-learn for Isolation Forest, TensorFlow/Keras for LSTM, XGBoost for compliance estimation, NetworkX for graph routing and a cryptography library for certificate and signature operations. Component-level implementation and integration status should be documented separately before submission.

Future work should validate the traffic generator, define and calibrate trust features, train and evaluate the prediction models on scenario-separated data, implement and profile genuinely distributed DPSO updates, and test the interaction between route replanning and local yielding under safety constraints. Subsequent work may examine realistic communication latency, packet loss, certificate revocation, privacy, scalability and transfer to real traffic data.

11 Conclusion

This paper has presented a decentralised emergency vehicle corridor coordination design that combines local digital twins, authenticated V2V message exchange, behavioural trust, anomaly screening, trajectory prediction, driver-compliance estimation, DPSO-based yielding coordination and A* route planning. The framework explicitly separates identity verification from information reliability, route selection from local movement coordination, and predicted compliance from observed driver response. A Python-based synthetic traffic environment provides the intended evaluation setting. Since validated experimental results were not available, the contribution is presented as a system

architecture and testable methodology; quantitative effectiveness remains future work.

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